

Abdul Samad Tariq

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Professional Summary

Results-driven Software Engineer with 3+ years of experience building end-to-end solutions across frontend, backend, and machine learning. Skilled at designing scalable systems, optimizing models, and delivering data-driven, user-facing applications. Comfortable adapting to whatever tools or frameworks a project demands, with a strong focus on clean code, efficient problem-solving, and practical, high-impact solutions.

Skills

- **Programming Languages:** Python, C++, C, Java, JavaScript, Assembly Language
- **Machine Learning:** Machine Learning Algorithms, Supervised & Unsupervised Machine Learning, Data Analysis, Data Warehousing, MLOps, Model Evaluation and Validation, Hyperparameter Tuning, Feature Engineering
- **Deep Learning:** Deep Neural Networks, Convolutional Neural Networks, Large Language Models (LLMs), Natural Language Processing (NLP), Generative Adversarial Networks (GANs), Recurrent Neural Networks (RNNs)
- **AI Development:** AI Ethics, AI Model Deployment, AI Chatbots, AI Agents & Assistants, Model Finetuning, Open Source LLMs, Vector Stores, Advance RAG Pipeline, AI Model Integration, OpenAI APIs
- **Libraries:** OpenCV, Pandas, Matplotlib, Seaborn, SciPy, NumPy, Word2Vec, Scikit-learn, Keras
- **Frontend Development:** HTML, CSS, JavaScript, Bootstrap, React.js, Tailwind CSS, Responsive Web Design
- **Backend Development:** Flask, Django, REST APIs, Web Sockets, Scalable System Design, System Administration
- **Databases:** MySQL, MongoDB, PostgreSQL
- **Deployment and UX:** Docker, Git and GitHub, Ubuntu, Linux (Shell Scripting)

Education

The Islamia University of Bahawalpur
Bachelor of Computer Science – CGPA: 3.4/4.0

2023 – 2027
Bahawalpur

- Relevant Coursework: Intro to Computing, Operating System, OOP, Computer Networking, DBMS, Artificial Intelligence, Data Science, Web Frameworks

Experience

Independent Developer
Self-Directed AI & Full-Stack Product Development

April 2026 – Present
Remote

- Built **MarketPulse**, an eBay marketplace intelligence platform (Django REST Framework, PostgreSQL, React/TypeScript), integrating eBay OAuth and the Browse API behind a marketplace abstraction layer, with an LLM layer returning structured JSON margin and opportunity analysis.
- Built **TechConnect AI**, a real-time professional networking platform (Flask, MongoDB, React) with JWT authentication, profile APIs, and Socket.IO-based user matching; designed WebRTC signaling and planned embedding-based semantic matching with Qdrant.
- Developed LLM prototypes covering prompt engineering, structured outputs, tool calling, RAG, and AI agents; experimented with quantization and QLoRA fine-tuning of open-source models in PyTorch.
- Published course-derived LLM and agentic AI projects as practical implementations for the developer learning community, iterating on them based on feedback.

Backend Engineer
Holistic TLC

February 2025 – March 2026
Bahawalpur

- Developed and maintained scalable backend applications using the Flask framework with MongoDB as the primary database, ensuring high performance and reliability under production workloads.
- Designed backend architecture with scalability as a core principle, implementing modular and maintainable code structures to support growing application demands.
- Led migration initiative from Flask to Django, demonstrating adaptability across Python web frameworks while maintaining continuity of backend services.
- Built and documented RESTful APIs to support frontend integration and third-party service communication, following best practices for security and performance.

Machine Learning Engineer
Enliven AI

May 2023 – Jan 2025
Bahawalpur

- Built an end-to-end ML solution for housing price prediction, integrating NLP embeddings and regression models (Linear Regression, Random Forest, DNN) with SHAP for interpretability, optimizing property listing success predictions.
- Developed an advanced unit conversion system using SmolLM2, implementing in-context learning, LoRA fine-tuning, and RFT to enhance reasoning, achieving high accuracy in generating precise conversion answers across diverse units.
- Designed a generalized text-to-speech voice cloning system using the YourTTS model, enabling multilingual speech synthesis from user-provided text and WAV files, optimized for Google Colab with GPU acceleration.